

$$Ex = \frac{\text{Exact} - \text{HPPook}}{Ex} = \frac{(1.1) - (1.2)}{(1.1)} = 8\%$$

⇒ Modified Euler Methods

⇒ Formula:

→ class work problem

$$y_{n+1} = y_0 + \frac{1}{2} h [f(n, y) + f(n_{n+1} + y'_n)]$$

⇒ ex:  $\frac{dy}{dn} = y - n, \quad y(0) = \frac{1}{2}$   
 $y = \frac{1}{2} \mid n=0$  where  $h=0.1$

$n=0$

$y_1 = \frac{1}{2} + 0.1 f(-n+y) \rightarrow$   $y_{n+1} = y_0 + h f(n, y)$  step 1 use Euler Method.

$$y_1 = \frac{1}{2} + 0.1 \left(-0 + \frac{1}{2}\right)$$

$$y_1 = \frac{1}{2} + 0.05$$

$$n_1 = h_1 + n_0 = 0 + 0.1 = 0.1$$

$$\boxed{y_1 = 0.55}$$

now step 2: Use MEuler Methods

$$y_1 = \frac{1}{2} + \frac{1}{2} (0.1) \left[ \frac{1}{2} + 0.1 + 0.55 \right]$$

$$\boxed{y_1 = 0.73} \text{ Ans}$$



ex2 solve

$\frac{dy}{dx} = -xy^2$ ,  $y=2$  at  $x=0$  by mod E method.  
y at  $x=0.2$  in two steps of 0.1 each

$$y_{n+1} = y_0 + hf(x_n, y_n)$$

$$y_{0+1} = y_0 + (0.1)(-xy^2)$$

$$y_1 = 2 + (0.1)(-0(-2)^2)$$

$y_1 = 2$

Now

$$y_1 = y_0 + \frac{1}{2} h [f(x_n, y_n) + f(x_{n+1}, y_{n+1})]$$

$x_1 = h + x_0$

$$y_1 = 2 + \frac{1}{2} (0.1) [0 + (0.1)(-2)^2]$$

$$y_1 = 2 + \frac{1}{2} 0.02 \quad \Bigg| \quad 2 + 0.05 [-0.1 \times (2)^2]$$

$$y_1 = 2 + 0.05$$

$$y_1 = 2.05$$

$$= 2 + 0.05 [-0.1 \times 4]$$

$$= 1.98$$